

topcatswap 1.1.1

User tutorial: desktop app and command-line bot

Swaps on Uniswap V2, V3 and V4 · Ethereum, Arbitrum and BSC

Ledger · Trezor · MetaMask · OKX · Rabby · Coinbase Wallet · Trust Wallet · imported keys

English (US) · Ubuntu

Contents

1. What topcatswap is
2. Choose your wallet
3. Ledger and Trezor
4. Browser wallets
5. Wallets imported by private key
6. Making a swap
7. Bundles
8. Wallets and balances
9. Settings and language
10. MEV protection (anti-sandwich)
11. Command-line bot
12. Troubleshooting

1 What topcatswap is

topcatswap compares the Uniswap pools (V2, V3 and V4) for a token pair, quotes each one with the real amount and picks the one that delivers the most. It then prepares, simulates and sends the swap, signed by the wallet you choose.

- Desktop app for Ubuntu (Swap, Pools, Wallets, Settings), in Portuguese or English.
- Command-line bot with POOL, SWAP and WALLET modes (topcatswap_comline_us_1-1-1.js and topcatswap_comline_pt_1-1-1.js).
- Protection against sandwich and frontrun: private sending (MEV Blocker, Flashbots; on BSC Merkle, PancakeSwap, 48 Club) with a new quote before signing.
- On-chain simulation before every send: if it fails, nothing is sent.
- Bundles tab (new in 1.1.1): several transactions in a single block, sent straight to the builders — all or nothing, private and no gas if it fails.

2 Choose your wallet

Every time the app opens it shows the wallet menu. The last one used is preselected; the language (PT | EN) is at the top right. To change later, use “Change wallet” on the card at the bottom left.

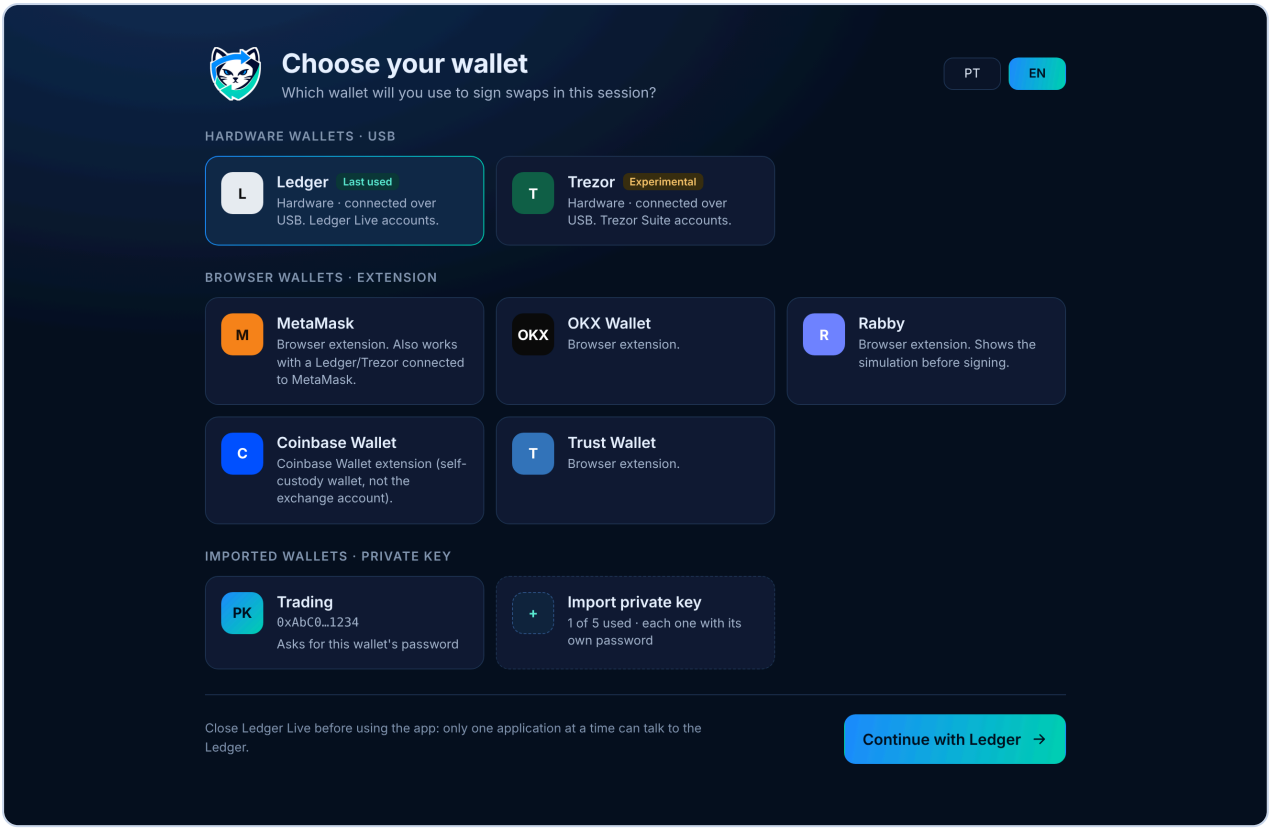


Figure 1. Wallet menu at startup.

Type	Wallets	How it signs
Hardware (USB)	Ledger, Trezor (experimental)	On the device screen; the key never leaves it.
Browser (extension)	MetaMask, OKX Wallet, Rabby, Coinbase Wallet, Trust Wallet	In the extension window, through a local page.
Imported	Up to 5 private keys	In the app, after unlocking it with that wallet's password.

3 Ledger and Trezor

Ledger

- udev rules (new PC): wget -q -O - https://raw.githubusercontent.com/LedgerHQ/udev-rules/master/add_udev_rules.sh | sudo bash
- Close Ledger Live: only one application at a time can talk to the Ledger.
- Ledger unlocked, Ethereum app open and “Blind signing” enabled (Ethereum app → Settings).

Trezor (experimental)

- udev rules: sudo curl -fsSL https://data.trezor.io/udev/51-trezor.rules -o /etc/udev/rules.d/51-trezor.rules then sudo udevadm control --reload-rules && sudo udevadm trigger
- Close Trezor Suite before using the app.
- Trezor One: the PIN is typed in the app on a 3×3 grid (the numbers are scrambled on the Trezor screen). Model T and Safe: PIN and passphrase on the device itself.
- Accounts m/44'/60'/0'/0/N, the same as Trezor Suite. Do a very small swap first.

4 Browser wallets

MetaMask, OKX Wallet, Rabby, Coinbase Wallet and Trust Wallet live as browser extensions. The app opens a local page (http://127.0.0.1:47615/..., only reachable from this computer) that connects the extension to the app.

1. Choose the wallet in the menu and click Continue: the page opens in the default browser.
2. On the page, click “Connect” and approve in the extension.
3. If the extension is in another browser, use “Copy page address” and paste it there.
4. Keep the page open while you use the app (you can minimize the browser).

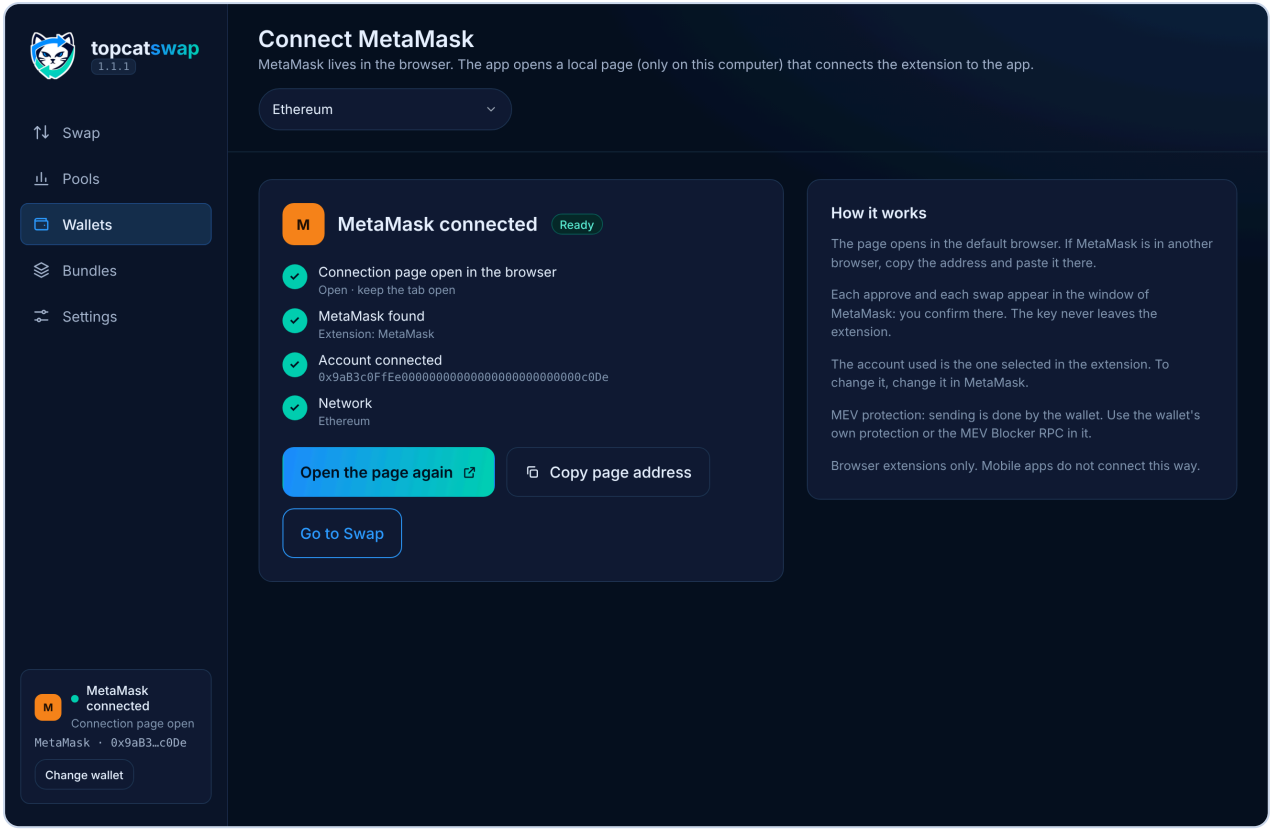


Figure 2. “Connect MetaMask” screen with the page open and the account connected.

> With these wallets the wallet itself sends the transaction, so topcatswap's MEV protection does not apply. Use the wallet's own protection (MetaMask: Smart Transactions) or add the RPC https://rpc.mevblocker.io to it. Mobile apps do not connect this way.

5 Wallets imported by private key

You can import up to 5 wallets by private key. Each one is encrypted with its own password (standard Ethereum keystore format, scrypt) in ~/.config/TopCatSwap/wallets/, readable only by your user. The plain key is never written to disk.

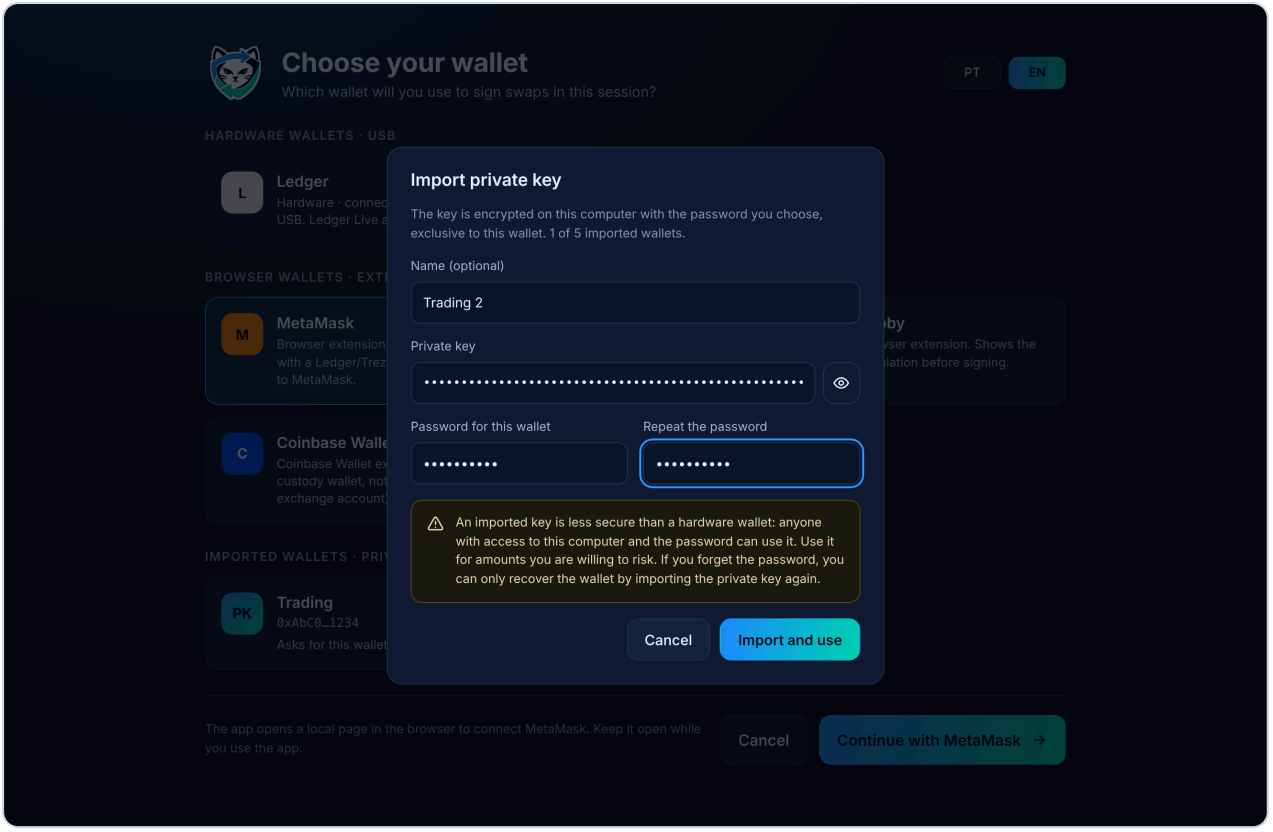


Figure 3. Import private key: name, key and a password exclusive to this wallet.

- When you choose an imported wallet, the app asks for that wallet's password. Ledger, Trezor and extensions do not ask for a password.
- It stays unlocked until you close the app, click “Lock”, choose another wallet or 30 minutes pass without use.
- Signing happens in the app with no external confirmation: review the summary carefully. MEV protection works normally.
- To import a sixth one, remove one in Settings › Imported wallets.
- The same wallets work with the command-line bot (--wallet=NAME).

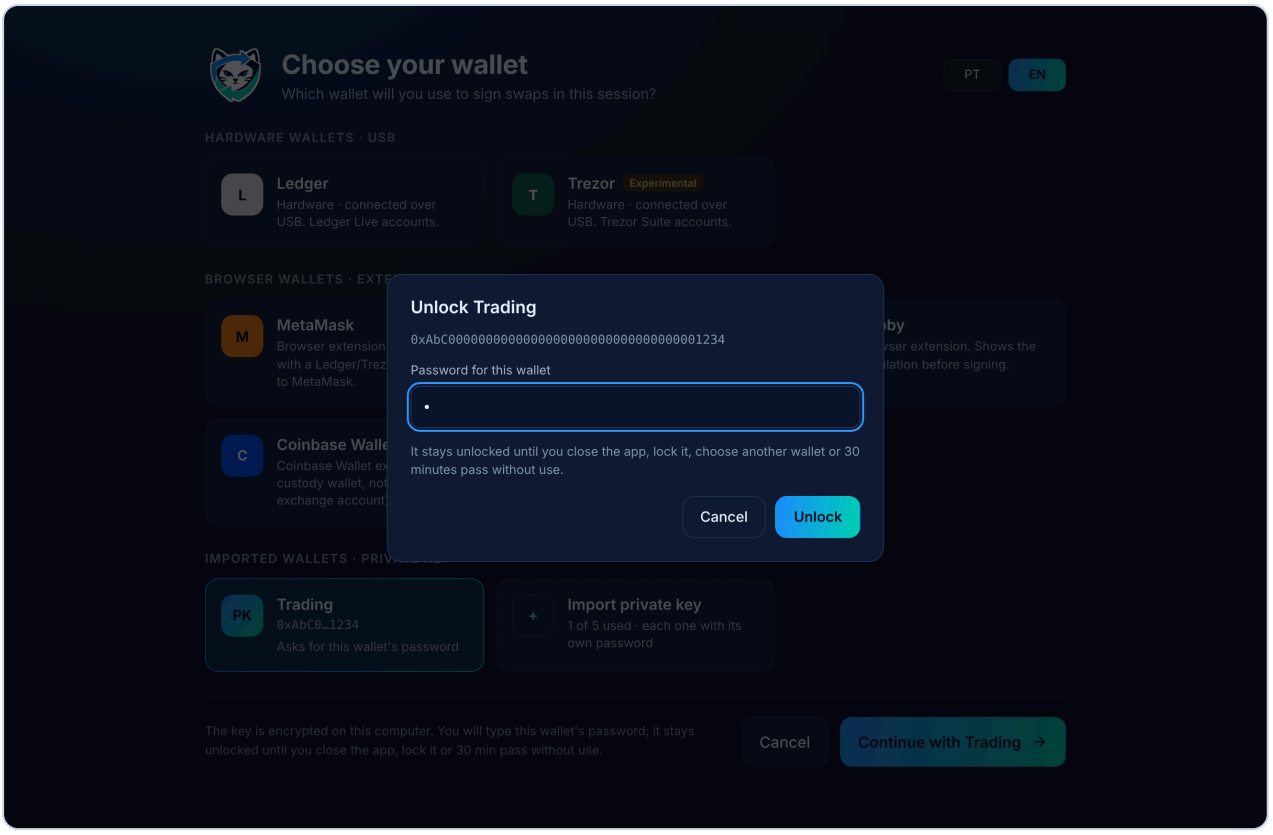


Figure 4. Unlocking an imported wallet with its password.

> An imported key is less secure than a hardware wallet: anyone with access to the computer and the password can use it. Keep the original key somewhere safe: if you forget the password, the only way back is to remove the wallet and import it again.

6 Making a swap

1. Swap: choose the token to sell and to receive (ETH, WETH, USDC or a contract address) and the amount.
2. Options: MEV protection, maximum slippage (presets or custom from 0.01 to 50%), speed and Uniswap versions.
3. Click “Analyze pools”. You can cancel the analysis at any time.
4. Pick the pool (the best execution is preselected) and click “Continue”.
5. Review the summary: guaranteed minimum, pool, wallet, sending and deadline. Click “Confirm and sign”.
6. Confirm in the wallet. The app shows the progress: approvals, new quote, simulation, signature, sending and inclusion in a block.

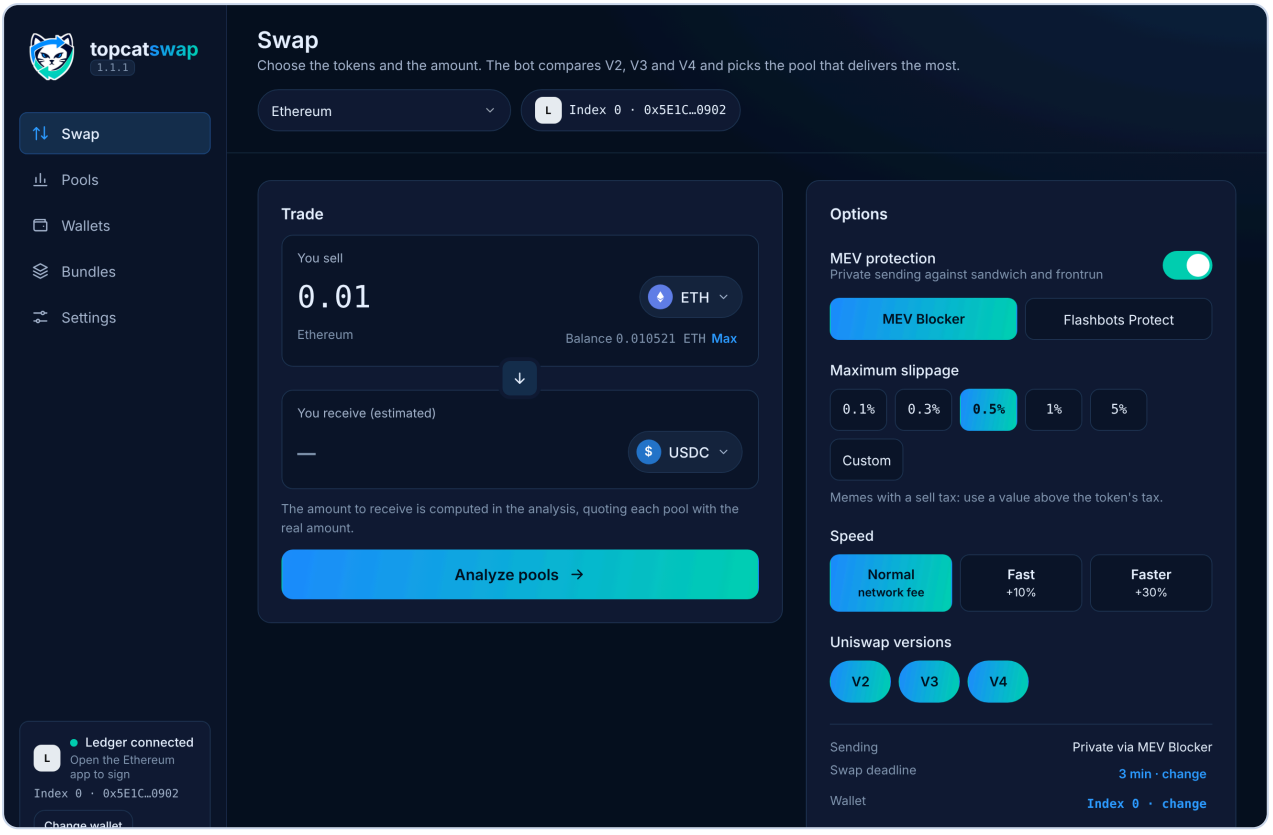


Figure 5. Swap screen with the options.

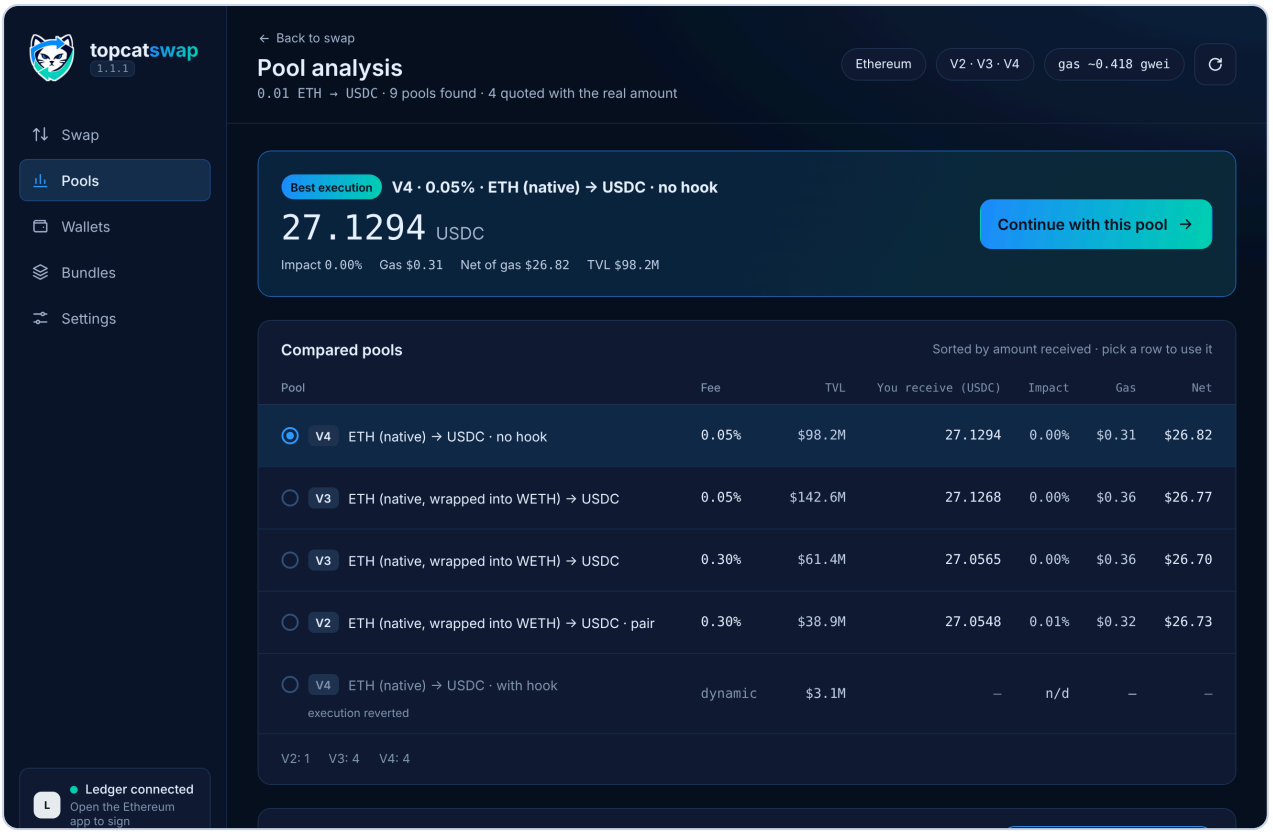


Figure 6. Pool analysis: best execution and the V2/V3/V4 comparison.

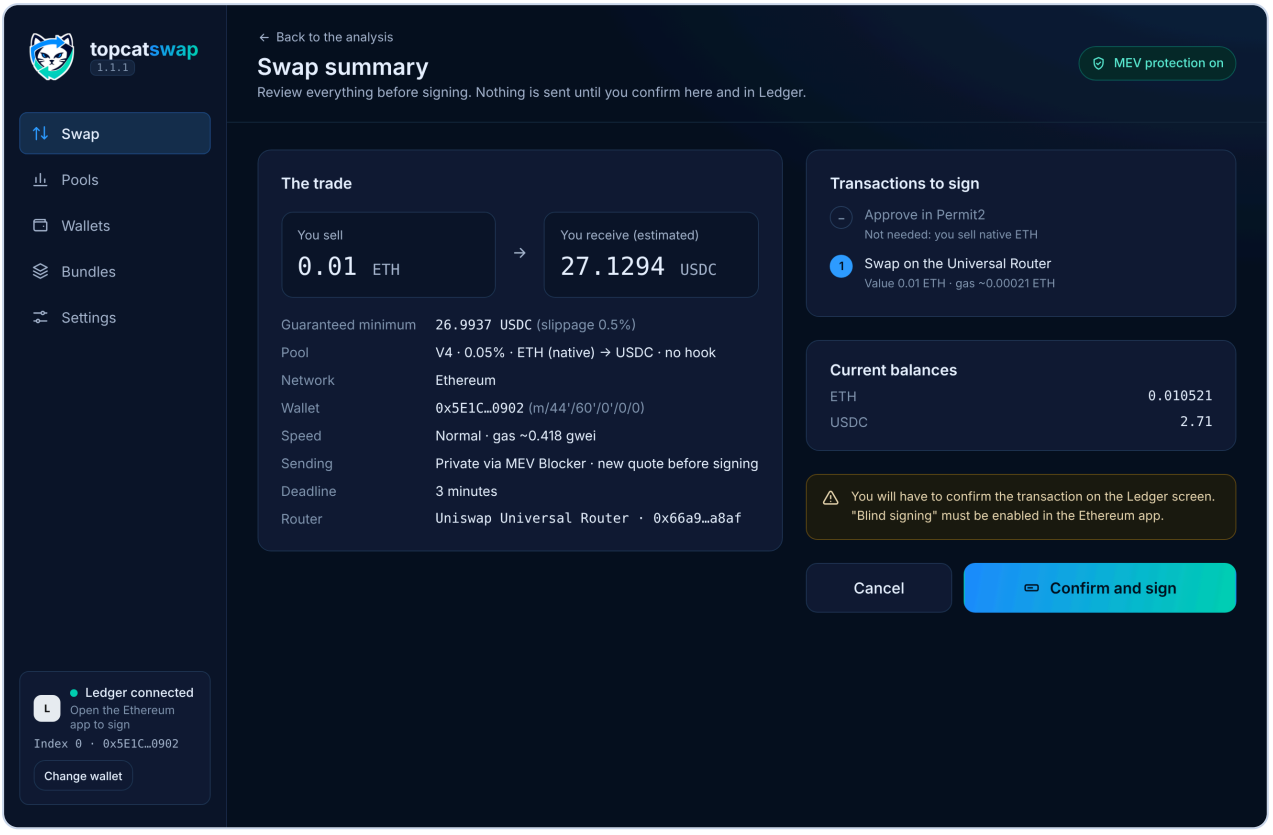


Figure 7. Swap summary before signing.

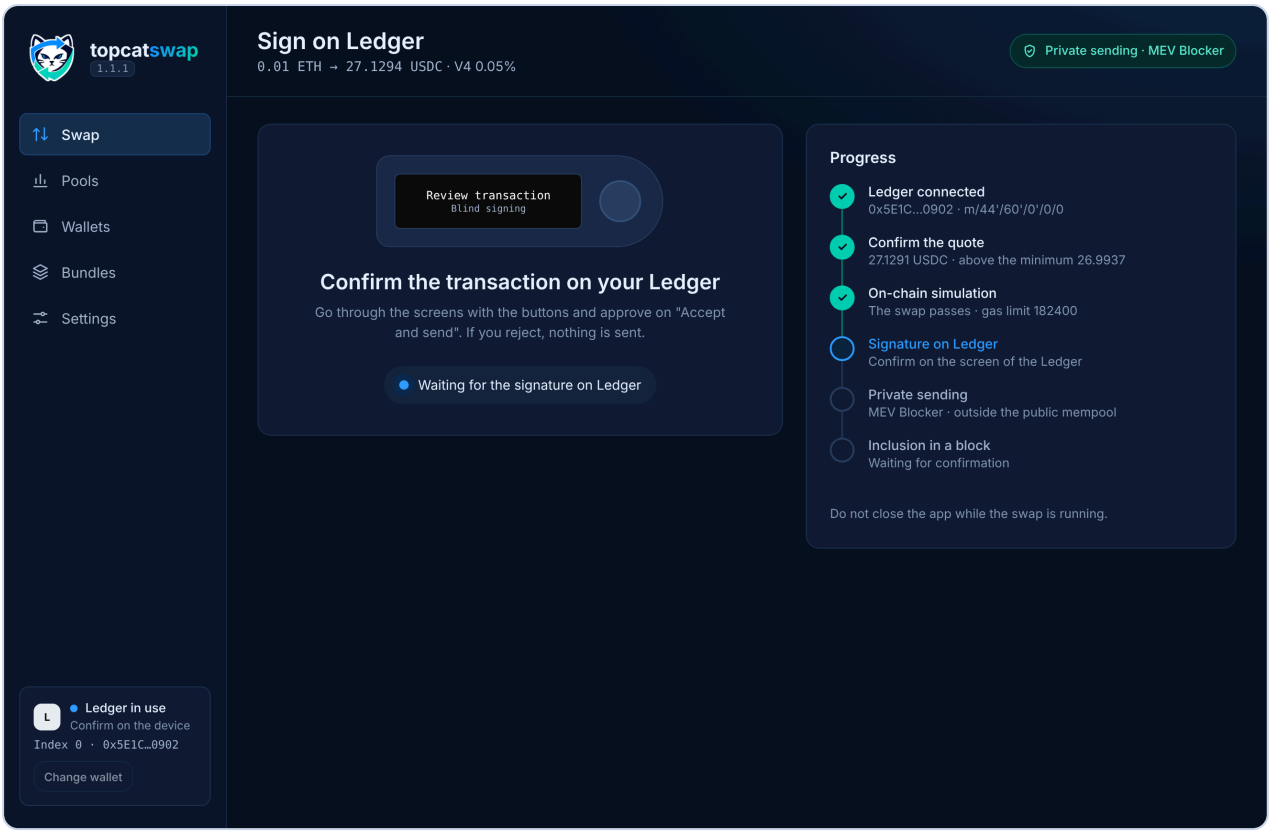


Figure 8. Signing and progress.

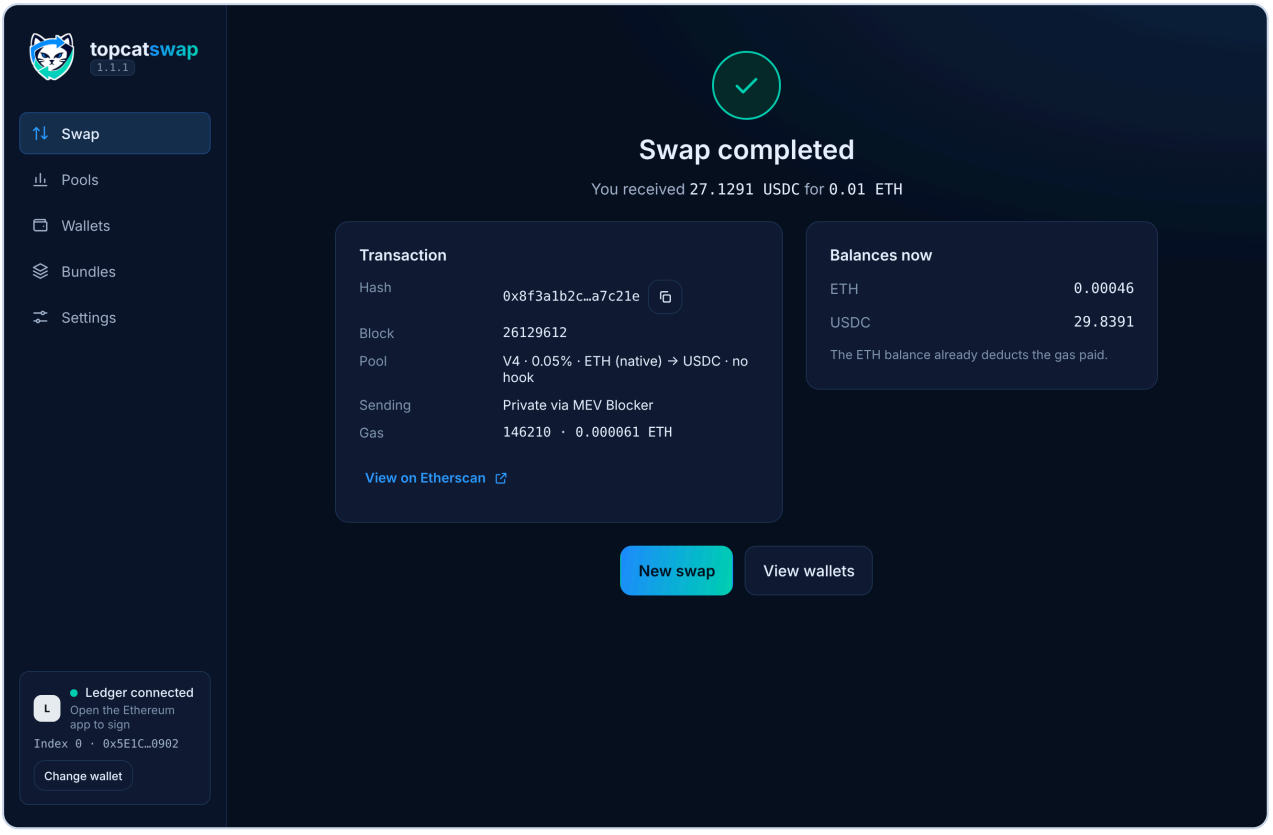


Figure 9. Swap completed, with the hash and balances.

> Do your first swap with a small amount. Tokens with a transfer tax (many memes): use V2 only and a slippage above the tax.

7 Bundles

The Bundles tab groups several transactions into a single block and sends them straight to the builders (who assemble the blocks), instead of the public mempool.

Advantage	What it means
All or nothing	The transactions all land in the same block, in the right order, or none does.
No gas if it fails	The builder simulates the bundle and does not include it if any transaction fails: you pay nothing.
Private	It never goes through the public mempool: no sandwich, no frontrun.
Several builders	Ethereum: Flashbots, BuilderNet, Titan, beaverbuild. BSC: 48 Club, BlockRazor. Sent over several blocks.

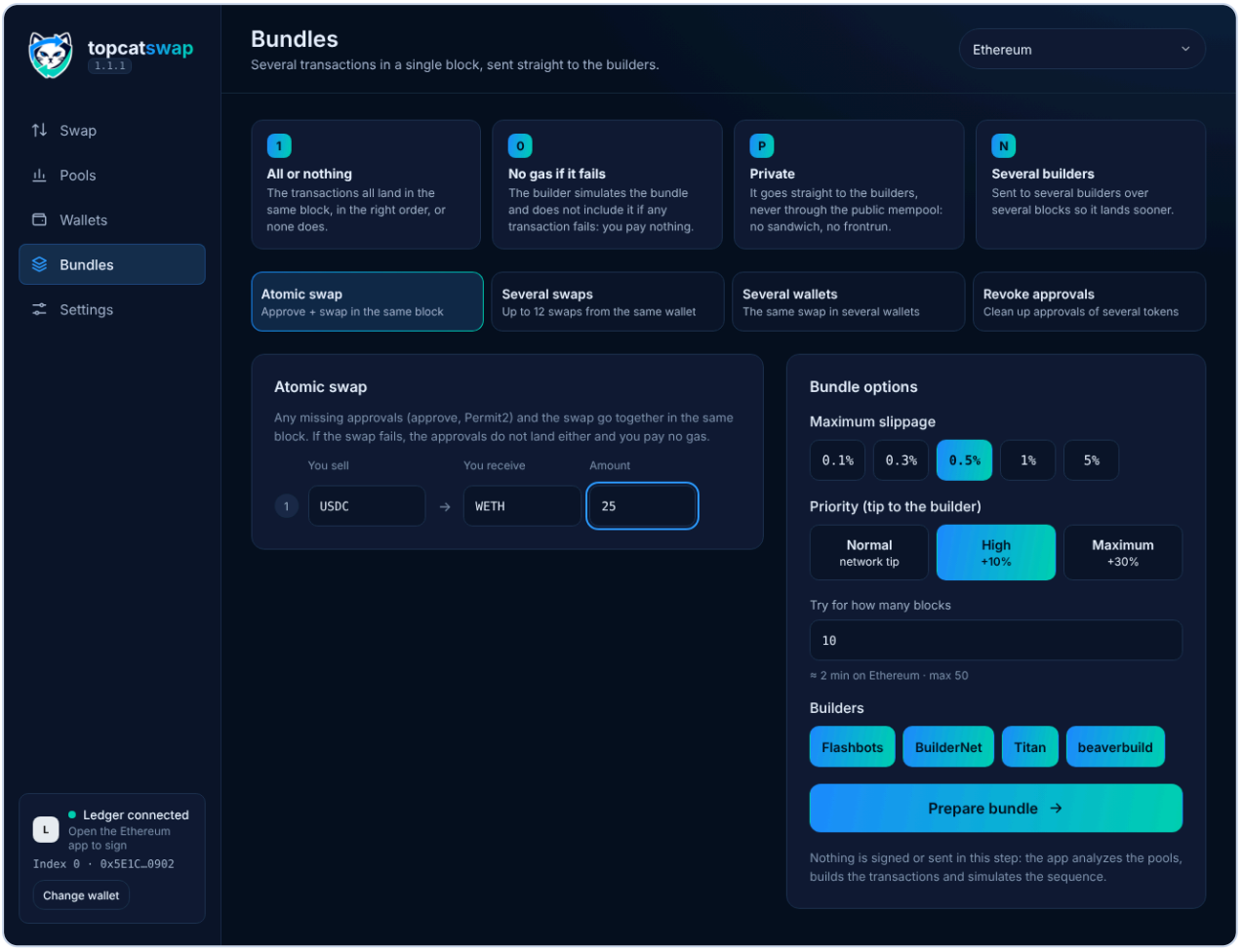


Figure 10. Bundles tab: advantages, bundle types and options.

The four types

- Atomic swap: any missing approvals (approve, Permit2) and the swap in the same block. If the swap fails, the approvals do not land either.
- Several swaps: up to 12 swaps from the active wallet in one block (e.g. sell several tokens at once).
- Several wallets: the same swap made by several Ledger/Trezor accounts or several imported wallets (each asks for its own password).

- Revoke approvals: shows and removes, in one bundle, the approvals given to Permit2 and the Universal Router.

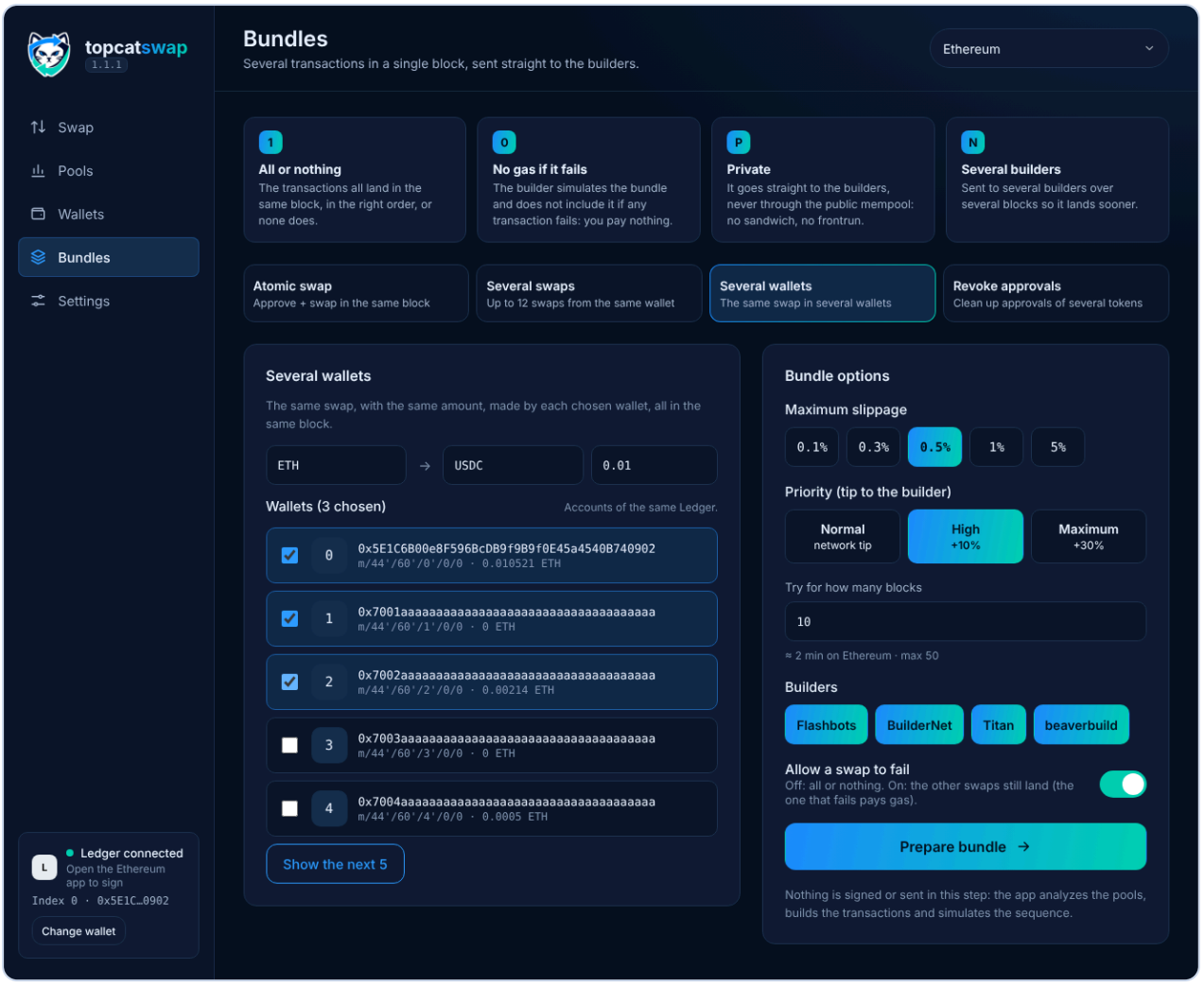


Figure 11. Several wallets: the same swap on three Ledger accounts.

Options

- Maximum slippage (swaps).
- Priority: the tip paid to the builder (Normal, High +10%, Maximum +30%). A higher tip = better chance to land quickly.
- Blocks: for how many blocks the bundle is tried (default 10 on Ethereum ≈ 2 min, 40 on BSC).
- Builders: where to send it.
- Allow a swap to fail (several swaps / several wallets): the others still land; the one that fails pays gas.

Step by step

1. Choose the type, fill in the swaps (or the tokens to revoke) and the options.
2. Click “Prepare bundle”: the app analyzes the pools and simulates the whole sequence without signing anything.
3. Review the summary: swaps, transactions, gas, block window and each wallet's balances.
4. Click “Sign and send bundle”. On the Ledger/Trezor you confirm each transaction; nothing is sent before you sign them all.

5. On Ethereum there is a final simulation at Flashbots; then the bundle goes to the builders and the app waits for the block.

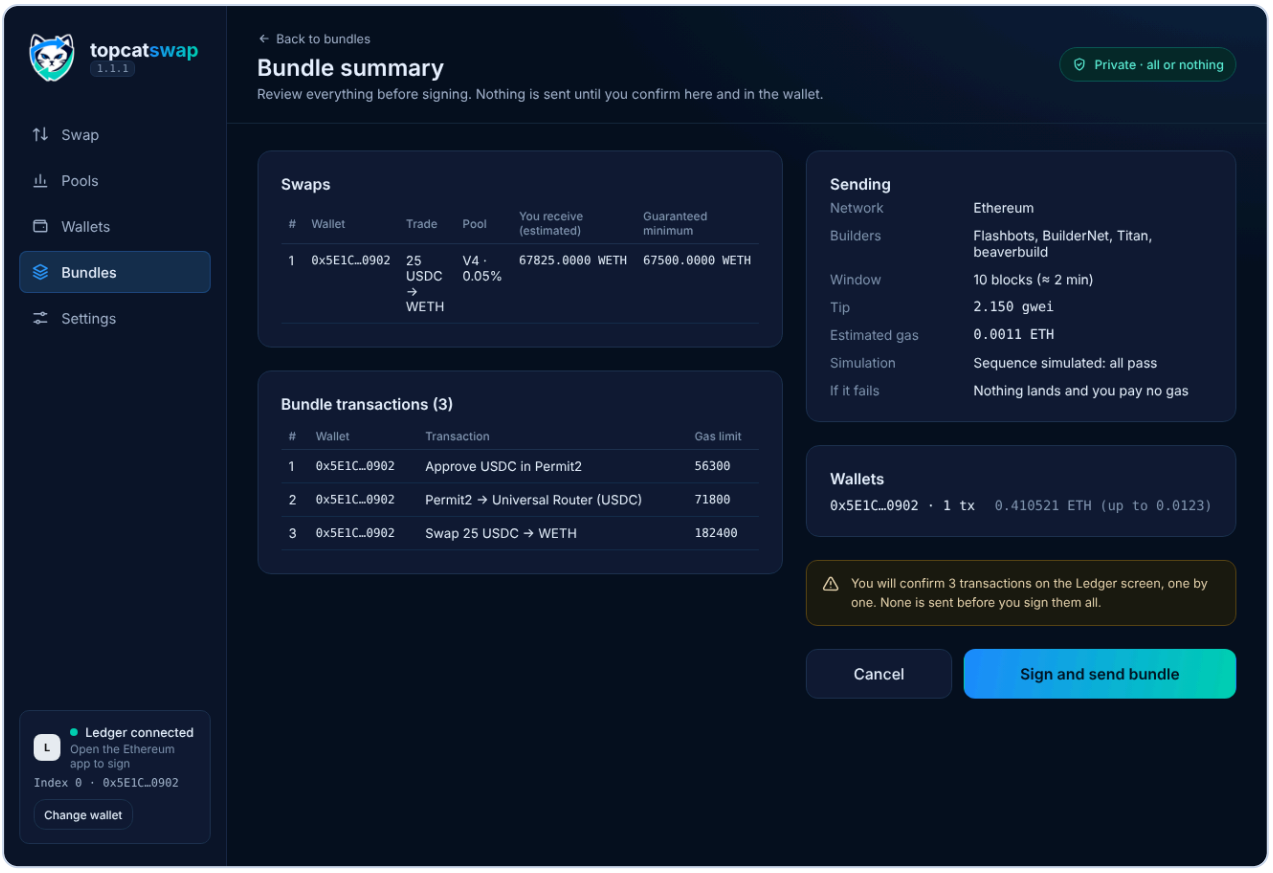


Figure 12. Bundle summary before signing.

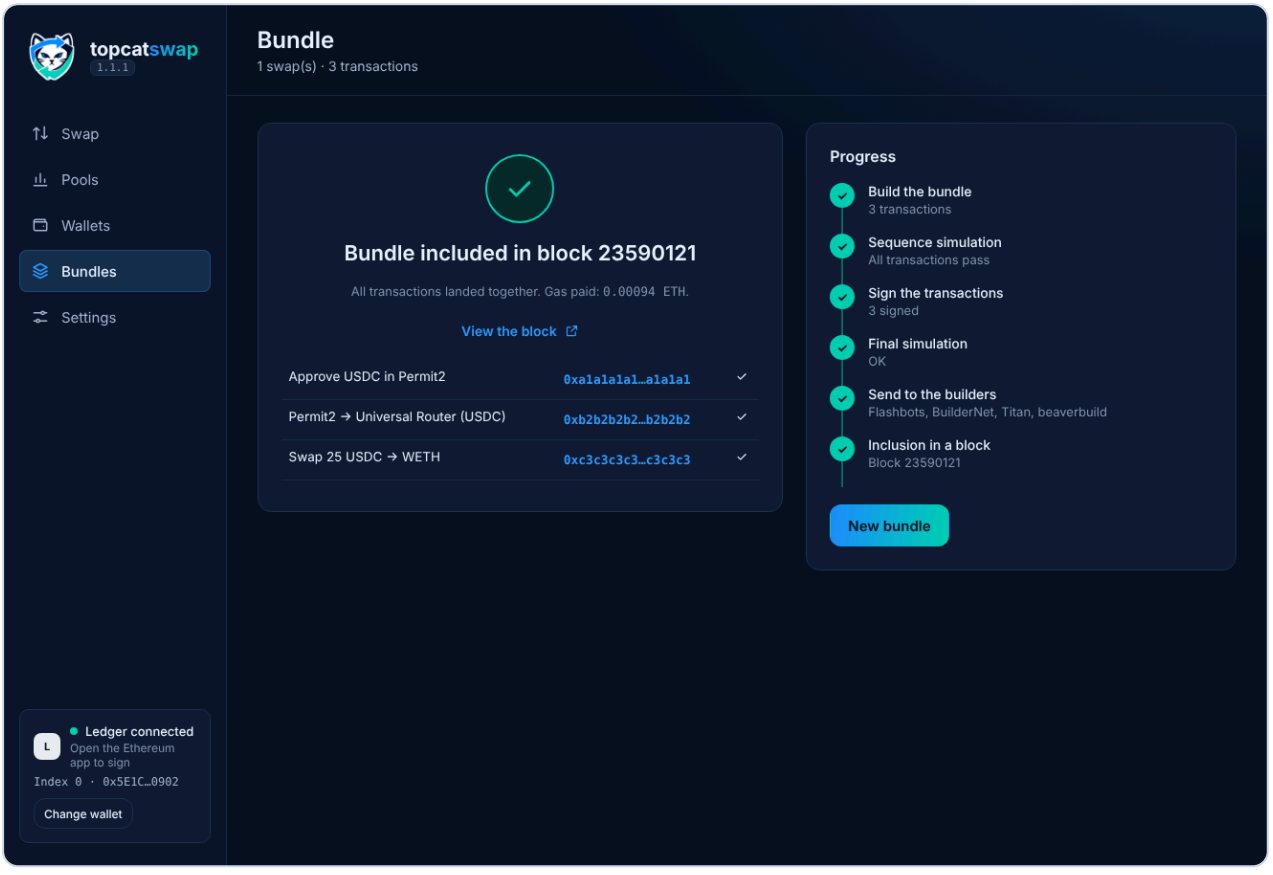


Figure 13. Bundle included: all transactions in the same block.

If no builder includes the bundle within the block window, nothing is executed and you pay no gas: try again with higher priority or more blocks.

Bundles only on Ethereum and BSC, with Ledger, Trezor or imported wallets. Browser extensions (MetaMask, OKX...) cannot sign without sending. Arbitrum has no bundles. BlockRazor may require an access key; in that case the bundle goes through 48 Club.

8 Wallets and balances

The Wallets screen shows the accounts and balances (ETH/BNB, USDC or any token by contract address). Ledger and Trezor show 5 accounts at a time (“Show the next 5”); “Use this one” sets the active wallet. With extensions, the active account is the one selected in the extension.

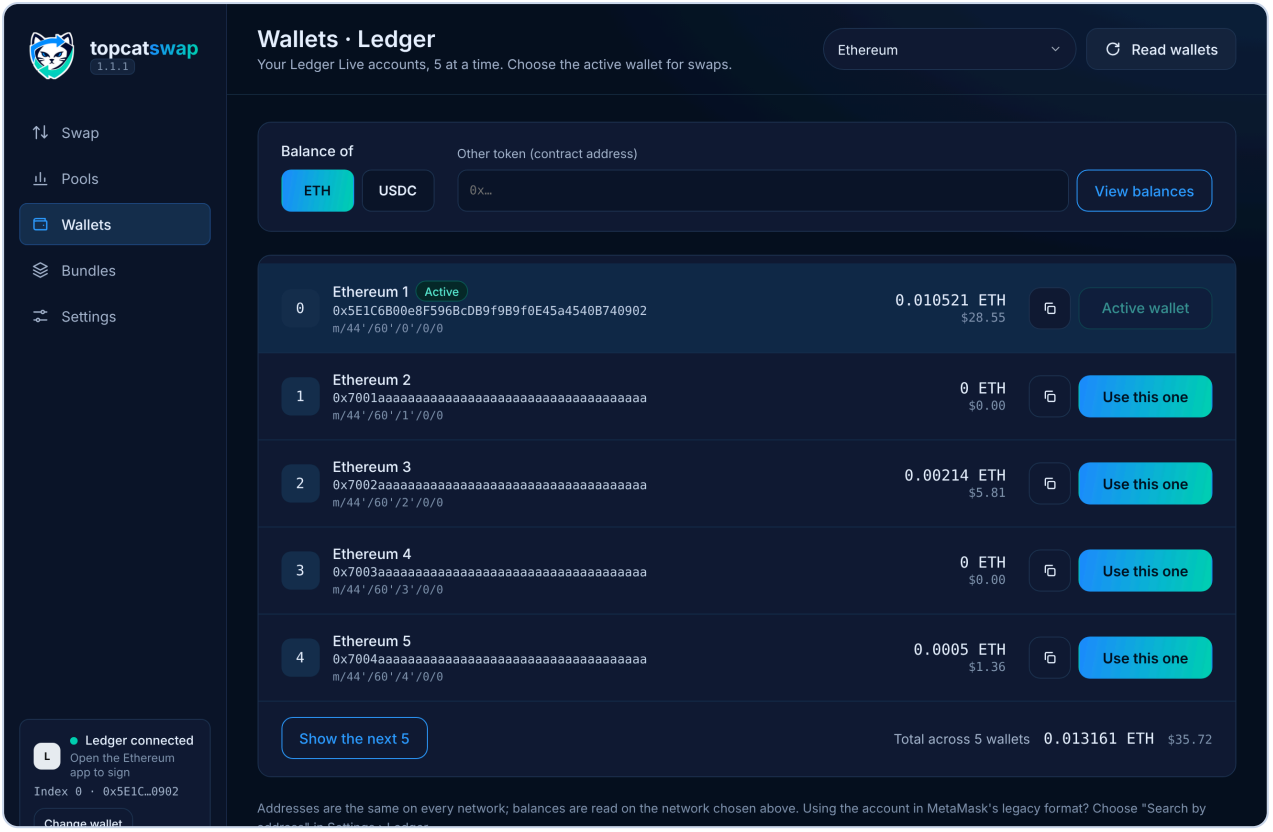


Figure 14. Ledger accounts with balances.

9 Settings and language

In Settings you choose the default network, slippage, speed, swap deadline (1 to 60 minutes, default 3), MEV protection and private sending service, RPCs, The Graph key, wallet and language.

Language: Português (PT) or English (US). The app restarts to change the language; settings and wallets are kept.

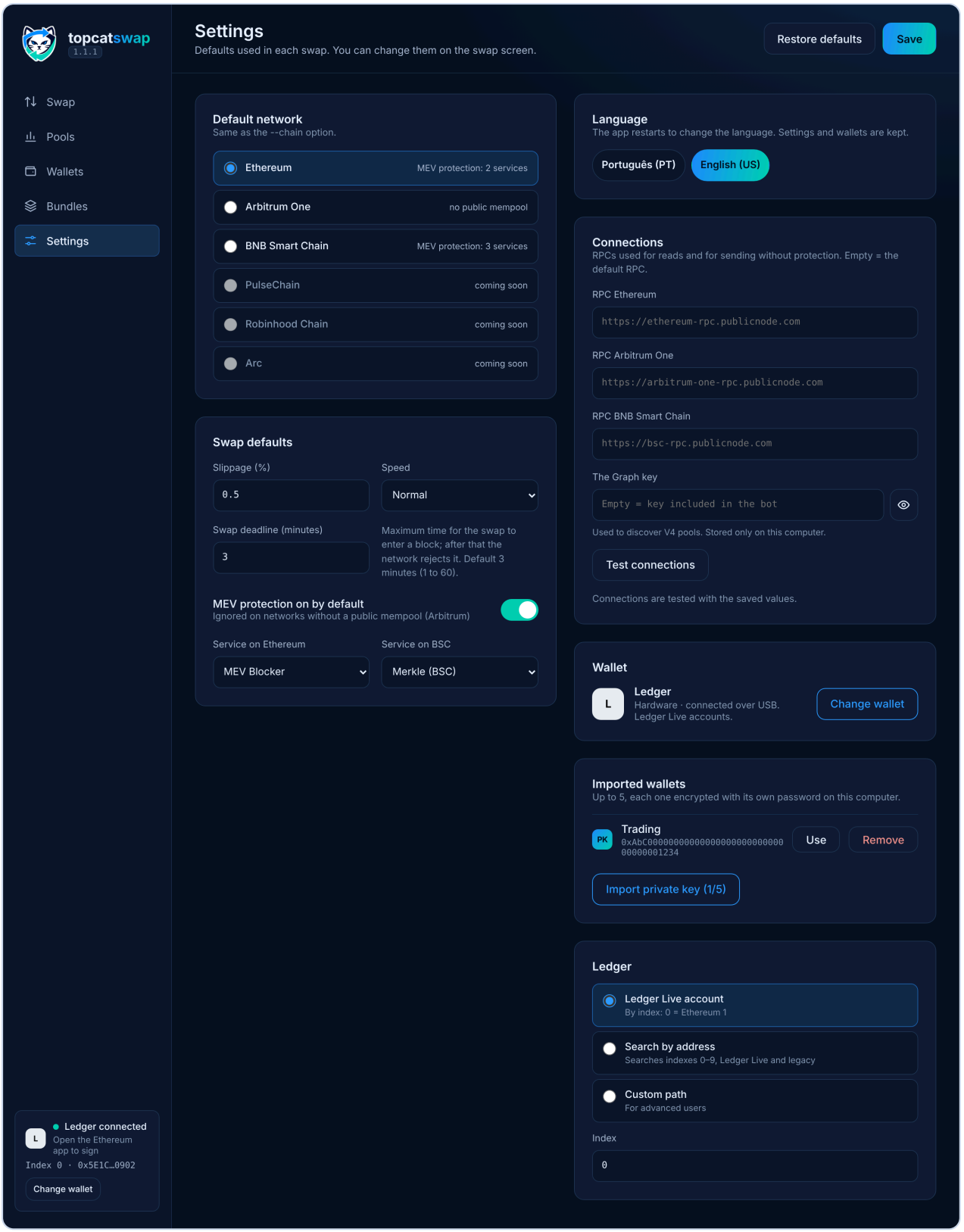


Figure 15. Settings, with the language and the imported wallets.

10 MEV protection (anti-sandwich)

With protection on, the transaction goes straight to a private RPC and never enters the public mempool, where sandwich bots would see it. Before signing, the app quotes the pool again: if the price got worse than the guaranteed minimum, nothing is sent.

Network	Services	Notes
Ethereum	MEV Blocker (default), Flashbots Protect	Private sending + new quote.
BNB Smart Chain	Merkle (default), PancakeSwap, 48 Club	Private sending + new quote.
Arbitrum One	—	No public mempool: the sequencer orders by arrival.

Slippage is still the final protection: it caps the maximum loss even without private sending.

11 Command-line bot

The same engine as the app, in a single file: topcatswap_comline_us_1-1-1.js (English) or topcatswap_comline_pt_1-1-1.js (Portuguese). Install the dependencies in the folder where you put it:

```
npm install ethers @ledgerhq/hw-transport-node-hid @ledgerhq/hw-app-eth
node topcatswap_comline_us_1-1-1.js            # full help
node topcatswap_comline_us_1-1-1.js --version
```

POOL mode (read-only)

```
node topcatswap_comline_us_1-1-1.js POOL ETH USDC
node topcatswap_comline_us_1-1-1.js POOL ETH USDC 10
node topcatswap_comline_us_1-1-1.js POOL MEME_CA ETH 1000000 --versions=v2
```

SWAP mode

```
node topcatswap_comline_us_1-1-1.js SWAP ETH USDC 0.01 --protect
node topcatswap_comline_us_1-1-1.js SWAP ETH USDC 0.01 --index=1 --fee=2
node topcatswap_comline_us_1-1-1.js SWAP ETH USDC 0.01 --wallet=Trading --protect
node topcatswap_comline_us_1-1-1.js SWAP BNB USDC 0.1 --chain=3 --protect
```

It shows a summary and only continues if you type YES. Options: --slippage=0.01–50, --fee=1|2|3 (normal, +10%, +30%), --protect[=service], --chain=1..6, --versions=v2,v3,v4, --index/--from/--path (Ledger), --wallet=NAME (imported wallet).

WALLET mode

```
node topcatswap_comline_us_1-1-1.js WALLET
node topcatswap_comline_us_1-1-1.js WALLET BALANCE USDC
node topcatswap_comline_us_1-1-1.js WALLET IMPORT
node topcatswap_comline_us_1-1-1.js WALLET IMPORTED
node topcatswap_comline_us_1-1-1.js WALLET REMOVE NAME
node topcatswap_comline_us_1-1-1.js WALLET BALANCE USDC --wallet=imported
```

The key and password are asked for without showing on screen. Imported wallets are the same as the app's (~/.config/TopCatSwap/wallets).

--chain	Network	Status
1	Ethereum	Supported
2	Arbitrum One	Supported (no public mempool)
3	BNB Smart Chain	Supported
4, 5, 6	PulseChain, Robinhood Chain, Arc	Not supported yet

Optional environment variables: RPC_URL, RPC_URL_42161, RPC_URL_56, GRAPH_API_KEY, TOPCATSWAP_WALLETS.

12 Troubleshooting

Message	What to do
Ledger not detected	Cable, PIN, udev rules; close Ledger Live.
Open the Ethereum app / Blind signing	Open the Ethereum app; enable Blind signing in Settings.
Trezor not detected	Trezor udev rules; close Trezor Suite.
Could not find MetaMask	The page opened in a browser without the extension: copy the address to the right browser.
The connection page is not open	“Open page” in the sidebar and keep the tab open.
Wrong password	Each imported wallet has its own password.
The swap simulation failed	The price moved or the token charges a tax: raise the slippage (memes: V2 only).
Build error	sudo apt install build-essential libudev-dev libusb-1.0-0-dev python3
Bundles not available	Switch to Ethereum or BSC and use a Ledger, Trezor or an imported wallet.
Bundle not included	Nothing was executed or paid: raise the priority or the blocks and try again.

The error log is at ~/.config/TopCatSwap/topcatswap.log (Ctrl+Shift+I opens the diagnostic tools).